

Experiment 3: Inventory Management Using Hooks

Aim

To create a product listing page using multiple stateless React components and demonstrate the use of props for passing data and handling user interactions.

Software Requirements

- Node.js
- ReactJS
- VS Code
- Web Browser

Procedure

1. Create a React application.
2. Create Header and ProductCard components.
3. Pass product details through props.
4. Display product name, price, and category.
5. Add a "Buy Now" button.
6. Run the application and verify the output.

Program (App.js)

```
import React, { useState, useMemo } from "react";
```

```
export default function App() {
```

```
  const [products] = useState([  
    { name: "Pen", qty: 10, price: 20 },  
    { name: "Book", qty: 5, price: 100 }  
  ]);
```

```
  const [search, setSearch] = useState("");
```

```
  const [dark, setDark] = useState(false);
```

```
  const totalValue = useMemo(() => {  
    return products.reduce(  
      (sum, item) => sum + item.qty * item.price,
```

```

    0
  );
}, [products]);

const filtered = products.filter((p) =>
  p.name.toLowerCase().includes(search.toLowerCase())
);

return (
  <div
    style={{
      backgroundColor: dark ? "#222" : "#fff",
      color: dark ? "#fff" : "#000",
      minHeight: "100vh",
      padding: "20px"
    }}
  >
    <h2>Inventory Management</h2>

    <button onClick={() => setDark(!dark)}>
      Toggle Theme
    </button>

    <br /><br />

    <input
      type="text"
      placeholder="Search Product"
      value={search}
      onChange={(e) => setSearch(e.target.value)}
    />

```

```
<ul>
  {filtered.map((p, index) => (
    <li key={index}>
      {p.name} - Qty: {p.qty} - ₹{p.price}
    </li>
  ))}
</ul>
```

```
<h3>Total Value: ₹{totalValue}</h3>
```

```
</div>
```

```
);
```

Input – Output

Test Case 1

Input:

name="Laptop"

price="50000"

category="Electronics"

Output:

Laptop

Price: ₹50000

Category: Electronics

Buy Now

Test Case 2

Input:

name="Mobile"

price="20000"

category="Electronics"

Output:

Mobile

Price: ₹20000

Category: Electronics

Buy Now

Result

Thus, a product listing page using stateless React components was successfully created.